

Exercise II

Network estimation from time series data

PSYNETS

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Conceptual Questions

Question 1

What are key differences between between-person and within-person network structures in multilevel VAR models & why is it problematic to assume that between-person associations generalize to the within-person level (or vice versa)?

Question 2

Why might assuming stationarity in psychological time-series data be problematic? Can you think of a real-world example of nonstationarity, and how this might affect the interpretation of standard GVAR models?

$N > 1$ time series

Install packages

Make sure to install all required packages:

```
# dplyr
if (!requireNamespace("dplyr", quietly = TRUE)) {
  install.packages("dplyr")
}
library(dplyr)

# mlVAR
if (!requireNamespace("mlVAR", quietly = TRUE)) {
  install.packages("mlVAR")
}
library(mlVAR)

# psychometrics
if (!requireNamespace("psychometrics", quietly = TRUE)) {
  install.packages("psychometrics")
}
library(psychometrics)

# qgraph
if (!requireNamespace("qgraph", quietly = TRUE)) {
  install.packages("qgraph")
}
library(qgraph)

# mgm
if (!requireNamespace("mgm", quietly = TRUE)) {
  install.packages("mgm")
}
library(mgm)
```

This morning, we practiced with estimating $N = 1$ networks using data collected by Fried, Papanikolaou, and Epskamp (2020). In this part of the exercise we will use this dataset to practice with estimating a multilevel VAR

network (`mlVAR`). For that purpose, we will use the data of all people in the dataset in combination with the `mlVAR` package.

Download the file `clean_network.RData` from [OSF](#) and read it into R:

```
load("clean_network.RData")
```

Next, run the following code to improve the node labels and to remove some redundant items:

```
# Variables to investigate:
vars <- paste0("Q",1:18)

# Labels:
varLabs <- c("Relax","Irritable","Worry","Nervous","Future","Anhedonia",
             "Tired","Hungry","Alone","Angry","Social_offline","Social_online",
             "Music","Procrastinate","Outdoors","C19_occupied","C19_worry",
             "Home")

# Rename columns in data:
names(Data2)[names(Data2) %in% vars] <- varLabs

# Remove items:
Data2 <- Data2 %>% select(-Hungry,-Angry,-Music,-Procrastinate)
varLabs <- varLabs[!varLabs %in% c("Hungry","Angry","Music","Procrastinate")]
```

You now have two objects in your R environment: `Data2` containing the data, and `varLabs` containing the names of the variables to be included.

Multilevel VAR model

To estimate a multilevel VAR (`mlVAR`) model, we will use the entire cleaned dataset (`Data2`) in combination with the `mlVAR` package.

Question 3

Look at the help file for the `mlVAR` function and estimate a multilevel-VAR model on the entire dataset (i.e., include all participants) on 4 to 6 variables of your choice. Given what you know about the data (e.g., the number of variables), would you recommend using orthogonal or correlated estimation for `mlVAR`?

Question 4

Look at the `mlVAR` plot method help file (`?plot.mlVAR`), and plot the estimated fixed-effect *temporal*, *contemporaneous* and *between-subjects* networks. Plot all networks with the same layout (e.g., circle layout or average layout), and hide (threshold) edges that are not significant at $\alpha = 0.05$. For the contemporaneous and between-subjects networks, use an “and”-rule to minimize type-1 error rate in showing edges.

Question 5

Plot the estimated individual differences for both the *temporal* and the *contemporaneous* network (in this case you don’t need to fix both structures to have the same layout as we don’t want to compare the structures to one another directly). Think about what the edges in these networks represent. Inspect the plotted networks, what conclusion can you draw about the heterogeneity in this sample based on these network structures?

Tip: Take a look at [Bringmann et al., \(2013\)](#)

Panel VAR model

In this part of the exercise you will analyze data collected by McBride et al., (2020) during the COVID-19 pandemic. For a detailed description of the data collection and the variables measured in the full dataset have a look at their [article](#).

We will only use a subset of their [original open access dataset](#) to practice with estimating network models based on panel data. The dataset we will use consists of *detrended*, 6 wave panel data for $n = 677$ participants. As the data we work with for the exercise is detrended, it becomes difficult to interpret the raw data values. The items included in the dataset are the following:

Item	Abbreviation	Description
1	pid	ID of the participant
2	wave	time of measurement
		Over the last two weeks, how often have you been bothered by the following problems?
3	Dep1	Little interest or pleasure in doing things
4	Dep2	Feeling down, depressed, or hopeless
5	Dep3	Trouble falling or staying asleep, or sleeping too much
6	Dep4	Feeling tired or having little energy
7	Dep5	Poor appetite or overeating
8	Dep6	Feeling bad about yourself - or that you are a failure or have let yourself or your family Down
9	Dep7	Trouble concentrating on things, such as reading the newspaper or watching television
10	Dep8	Moving or speaking so slowly that other people have noticed? Or the opposite - being so fidgety or restless that you have been moving around more than usual
11	Dep9	Thoughts that you would be better dead or of hurting yourself in some way
12	DepTotal	Total depression score (sum of 9 'Dep' items)
13	GAD1	Feeling nervous, anxious or on edge
14	GAD2	Not being able to stop or control worrying
15	GAD3	Worrying too much about different things
16	GAD4	Trouble relaxing
17	GAD5	Being so restless that it is hard to sit still
18	GAD6	Becoming easily annoyed or irritable
19	GAD7	Feeling afraid as if something awful might happen
20	GADTotal	Total GAD score (sum of 7 'GAD' items)

Note: Range of all items 1-4: not at all, several days, more than half of the days, nearly every day.

Download the file `panel_data.RData` and read it into R:

```
load("panel_data.RData")
```

Inspect the datafile:

```
View(panel_data)
```

As you may notice, the format of this datafile looks different from the one we have worked with so far. When working with repeated-measures data, time-series data can be structured in either a *long* format or *wide* format. The previous datasets we used were in a *long* format, where each time point is specified in a separate column, and each measurement occupies its own row. This means that for a single individual, the datafile contains multiple rows corresponding to the number of assessments. In contrast, the current datafile is in a wide format. In a wide dataformat, there is no separate column to indicate the time point. Instead, each individual is represented by a single row, with separate columns corresponding to each time point.¹

Some functions require the data to be in a long format, while others work with data in a wide format. Therefore it is recommended to consult the helpfile of the function you plan to use to check the required data structure. In order to estimate a panel GVAR model using the `panelgvar` function from the `psychonetrics` package, the dataset must be in a wideformat.

In addition to a wide data format, if we review the documentation with `?panelgvar`, we also see that `vars` is a required argument, but needs to be specified differently when we have done before. Instead of a vector containing the names of the variables we want to include in our network model, `vars` now requires us to specify a matrix in which each row indicates a variable and each column indicates a measurement:

```
# Create a design matrix:
varnames <- colnames(panel_data[2:55])
designvars <- matrix(varnames, 9, 6)
designvars
```

¹You can reshape the data in R from one format into the other. A demonstration of this goes beyond the scope of this assignment.

```
##      [,1]      [,2]      [,3]      [,4]      [,5]      [,6]
## [1,] "W1_Dep_1" "W2_Dep_1" "W3_Dep_1" "W4_Dep_1" "W5_Dep_1" "W6_Dep_1"
## [2,] "W1_Dep_2" "W2_Dep_2" "W3_Dep_2" "W4_Dep_2" "W5_Dep_2" "W6_Dep_2"
## [3,] "W1_Dep_3" "W2_Dep_3" "W3_Dep_3" "W4_Dep_3" "W5_Dep_3" "W6_Dep_3"
## [4,] "W1_Dep_4" "W2_Dep_4" "W3_Dep_4" "W4_Dep_4" "W5_Dep_4" "W6_Dep_4"
## [5,] "W1_Dep_5" "W2_Dep_5" "W3_Dep_5" "W4_Dep_5" "W5_Dep_5" "W6_Dep_5"
## [6,] "W1_Dep_6" "W2_Dep_6" "W3_Dep_6" "W4_Dep_6" "W5_Dep_6" "W6_Dep_6"
## [7,] "W1_Dep_7" "W2_Dep_7" "W3_Dep_7" "W4_Dep_7" "W5_Dep_7" "W6_Dep_7"
## [8,] "W1_Dep_8" "W2_Dep_8" "W3_Dep_8" "W4_Dep_8" "W5_Dep_8" "W6_Dep_8"
## [9,] "W1_Dep_9" "W2_Dep_9" "W3_Dep_9" "W4_Dep_9" "W5_Dep_9" "W6_Dep_9"
```

Question 6

Estimate a pruned panel VAR model using the `panelgvar` function from the `psychometrics` package. After estimating a saturated panel GVAR model, prune the model at an alpha level of 0.05, and set `recursive` to `FALSE`. Look at the help file `?ml_gvar` and the slides to see how to specify the arguments.

Question 7

Plot the pruned *temporal*, *contemporaneous*, and the *between-subjects* networks for the panel VAR network.

Note, the focus for this question is on the correct estimation of the model, *not* on the interpretation of the specific parameters. Several reasons make the interpretation of the model parameters for now difficult: (1) data is detrended and it is unclear how this precisely affects model interpretation, (2) the software used to estimate the panel VAR model is beta software and is currently under development.

N = 1 time-series

Time-Varying VAR model

The models we have estimated so far assumed the data was stationarity. However, real-world data often exhibit time-varying dynamics, particularly in psychological or clinical settings. In this part of the exercise, we will explore how to estimate a time-varying VAR model, which can help us capture changes in the relationships between variables over time.

For this part of the exercise, we will use data from an extensive Experience Sampling Method (ESM) study of a single individual diagnosed with major depression. The dataset includes 51 variables, recorded across 1476 time points over a period of 238 consecutive days (Wicherts et al., 2016). These variables cover a wide range of experiences, including mood, self-esteem, social interactions, physical activity, life events, and symptoms of depression.

The data was collected using notifications sent to the individual's smartphone at 10 time points each day, with intervals that were pseudo-randomized, averaging 90 minutes between 7:30 AM and 10:30 PM. During the data collection period, the individual underwent a double-blind medication dose reduction. The study spans a baseline period, the dose reduction phase, and two post-assessment periods.

You will use the time-varying VAR model to explore how the network of variables evolves over time, particularly in response to the changes in medication. The data along with a detailed description can be found in [Kossakowski et al. \(2017\)](#), but it is also automatically loaded under the name `symptom_data` within the `mgm` package.

```
# First 6 rows of first 6 variables:
```

```
head(symptom_data$data)[, 1:6]
```

```
## mood_relaxed mood_down mood_irritat mood_satisfi mood_lonely mood_anxious
## 1          5         -1           1           5          -1          -1
## 2          4          0           3           3           0           0
## 3          4          0           2           3           0           0
## 4          4          0           1           4           0           0
## 5          4          0           2           4           0           0
## 6          5          0           1           4           0           0
```

```
# Time information:
```

```
head(symptom_data$data_time)
```

```
##      date dayno beepno beeptime resptime_s resptime_e  time_norm
## 1 13/08/12  226     1   08:58   08:58:56   09:00:15 0.000000000
## 2 14/08/12  227     5  14:32   14:32:09   14:33:25 0.005164874
## 3 14/08/12  227     6  16:17   16:17:13   16:23:16 0.005470574
## 4 14/08/12  227     8  18:04   18:04:10   18:06:29 0.005782097
```

```
## 5 14/08/12 227 9 20:57 20:58:23 21:00:18 0.006285774
## 6 14/08/12 227 10 21:54 21:54:15 21:56:05 0.006451726
```

The dataset is contained in the list object `symptom_data`; `symptom_data$data` contains the measurements and `symptom_data$data_time` contains information about when the measurements were taken. The additional list entries contain the variable names, the type of each variable, the number of levels (or categories) of each variable, and a group label for each variable.

In order to estimate the network, we only consider 14 of the 51 variables. Specifically, we take the subset of all 12 mood-related variables and the two categorical variables *Action* and *Who with*. *Action* indicates the activity the individual was performing since the last notification. And *Who with* indicates with whom the individual spent time together since the last notification. We can subset this data as follows:

```
symptom_data_short <- symptom_data
selection_ind <- c(which(symptom_data$groups == "Mood"),
                  which(symptom_data$colnames %in% c("Action", "Who with"))))

# Subset:
symptom_data_short$data <- symptom_data_short$data[,selection_ind]
symptom_data_short$type <- symptom_data_short$type[selection_ind]
symptom_data_short$level <- symptom_data_short$level[selection_ind]
symptom_data_short$colnames <- symptom_data_short$colnames[selection_ind]
symptom_data_short$groups <- symptom_data_short$groups[selection_ind]
```

You can find the type and the levels of the variables in the datalist (`symptom_data_short$type` and `symptom_data_short$level`).

Question 8

Use the function `tvmvar()` to fit a time-varying VAR model to the data. Specify `lambdaSel = "CV"` and `lambdaFolds = 10` to select the regularization parameter with 10-fold cross-validation. Make use of the helpfile `?tvmvar` and the [mgm paper](#) (Haslbeck & Waldorp, 2019), if necessary.

As with previous methods used for the estimation of networks based on time series data, to ensure that only time points are used for estimation for which we have also the previous time point, we provide the number of the notification (the “beep”) via the argument `beepvar` and the day of via the argument `dayvar`.

To estimate a time-varying VAR model we need to specify two additional arguments: first, the bandwidth parameter (argument `bandwidth`), which determines how many observations close in time are combined to estimate a model at a given estimation point. For this exercise choose `bandwidth = 0.15`. Second, the number and spacing of estimation points. For this exercise use `estpoints = seq(0, 1, length = 10)`, which estimates ten models at equally spaced time intervals across the entire time series.

Question 9

Visualize the time-varying VAR models at estimation point 3 and estimation point 8, side by side and make sure that the two plots are comparable by using the same layout. Remember, you previously specified a total of 10 estimation points. Use the `qgraph` package to plot the networks.

Make sure that you transpose the parameter matrix for both models (using `t()`), so they are correctly oriented for visualization in `qgraph` and provide the edge colors (which indicate the sign of the relationship) separately via the `qgraph` argument `edge.color`. You can find them in the `tmvar()` output object. You can find labels for the variables in `symptom_data_short$colnames`.

Tip: Both the parameter matrix and the matrix of edge colors need to be transposed (`t()`) before using them in `qgraph`. This is because in `mgm`, columns predict rows, whereas in `qgraph`, rows predict columns.